Dynamic Model — Dialog Map

SC2006 Team 4 · Dialysis Transport Coordination System · Lab 2 Deliverable

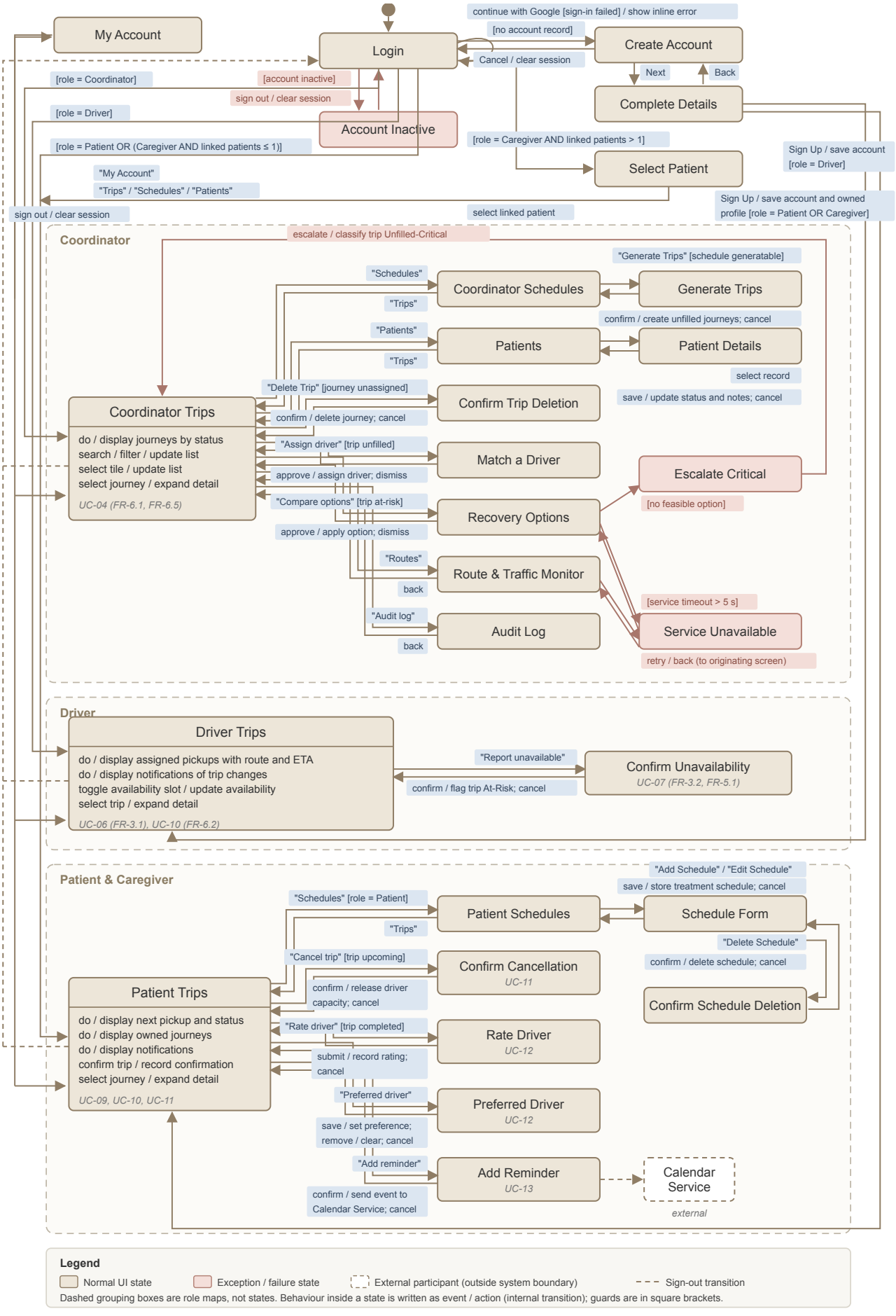
This document models the system's behaviour as a state machine of user interface screens, following Lab 2 §3.3.1. States are UI screens; transitions are user actions or system events. The model follows the routes and screens the system provides; states that are designed but have no data path behind them yet are identified in the design notes.

Modelling conventions

- A screen is a separate **state** only where the set of available actions changes. Filtering, sorting and expanding content in place are **internal transitions**, written inside the state box as `event / action`.
- `do /` denotes a continuous activity while the state is active. Event-driven behaviour that completes is written as an internal transition, not as `do /`.
- Guards are written in square brackets on the transition to which they apply.
- Exception and failure states are shaded differently from normal UI states.
- External systems outside the system boundary are drawn with a dashed outline and are not states.

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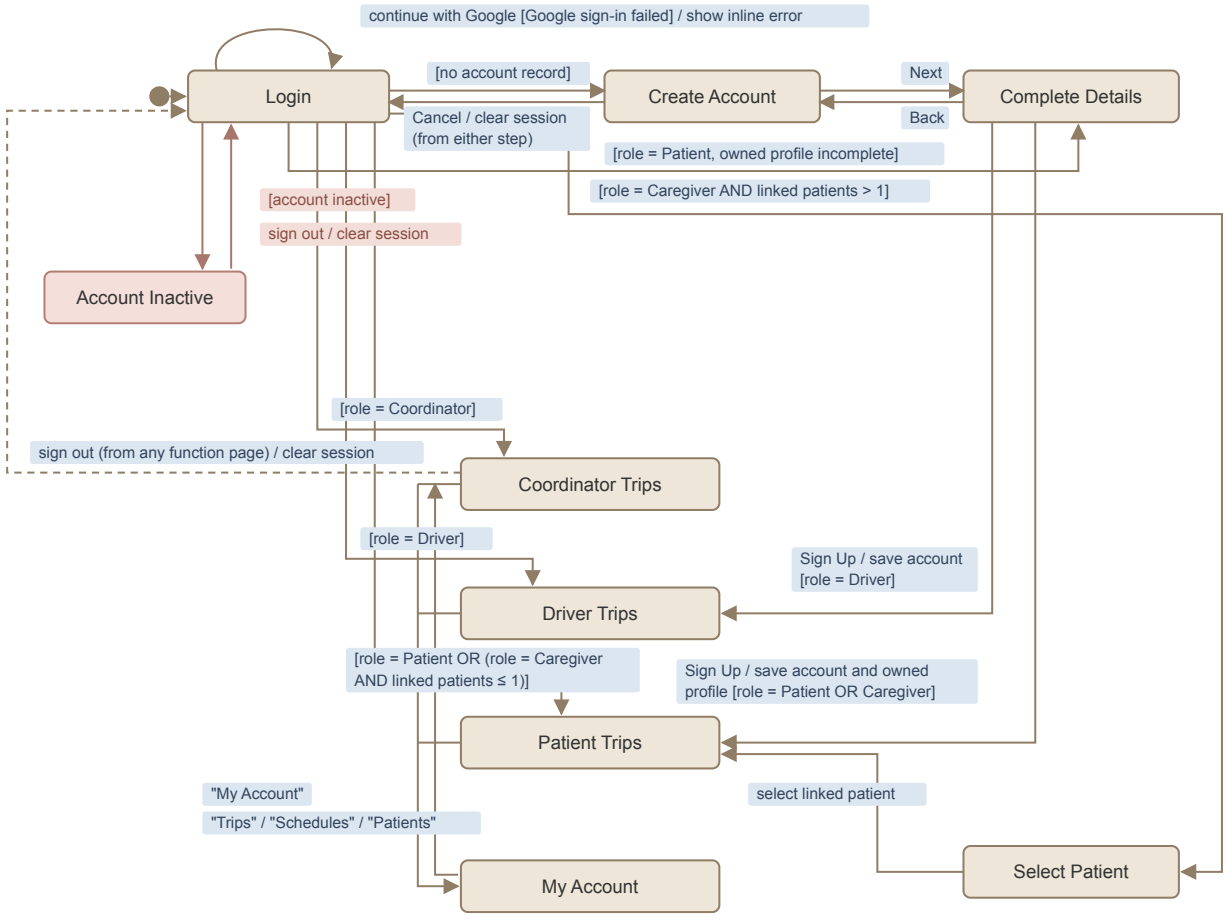
1. Complete Dialog Map



How to read this map

- This is the whole model in one diagram. Login, Create Account, Complete Details, Account Inactive, Select Patient and My Account belong to no role map and are shared by every role; the three dashed groupings are the Coordinator, Driver and Patient/Caregiver maps reached after authentication. The groupings are drawing conventions, not states.
- Sections 3–5 redraw each grouping on its own, at a larger size and with its own design notes. No state or transition appears in those sections that is not in this map, and none in this map is omitted from them. Login and My Account appear there as the shared targets they are; both are defined in Section 2.
- The guarded transitions out of Login determine which map the session enters. Each authenticated session is bound to a single role (FR-1.3), so only one grouping is reachable per session.
- Sign-out is drawn once, as the dashed line running from all three entry states back to Login, rather than repeated on every state.
- Section 6 lists every transition in tabular form and is the formal statement of the model.

2. Shared Entry



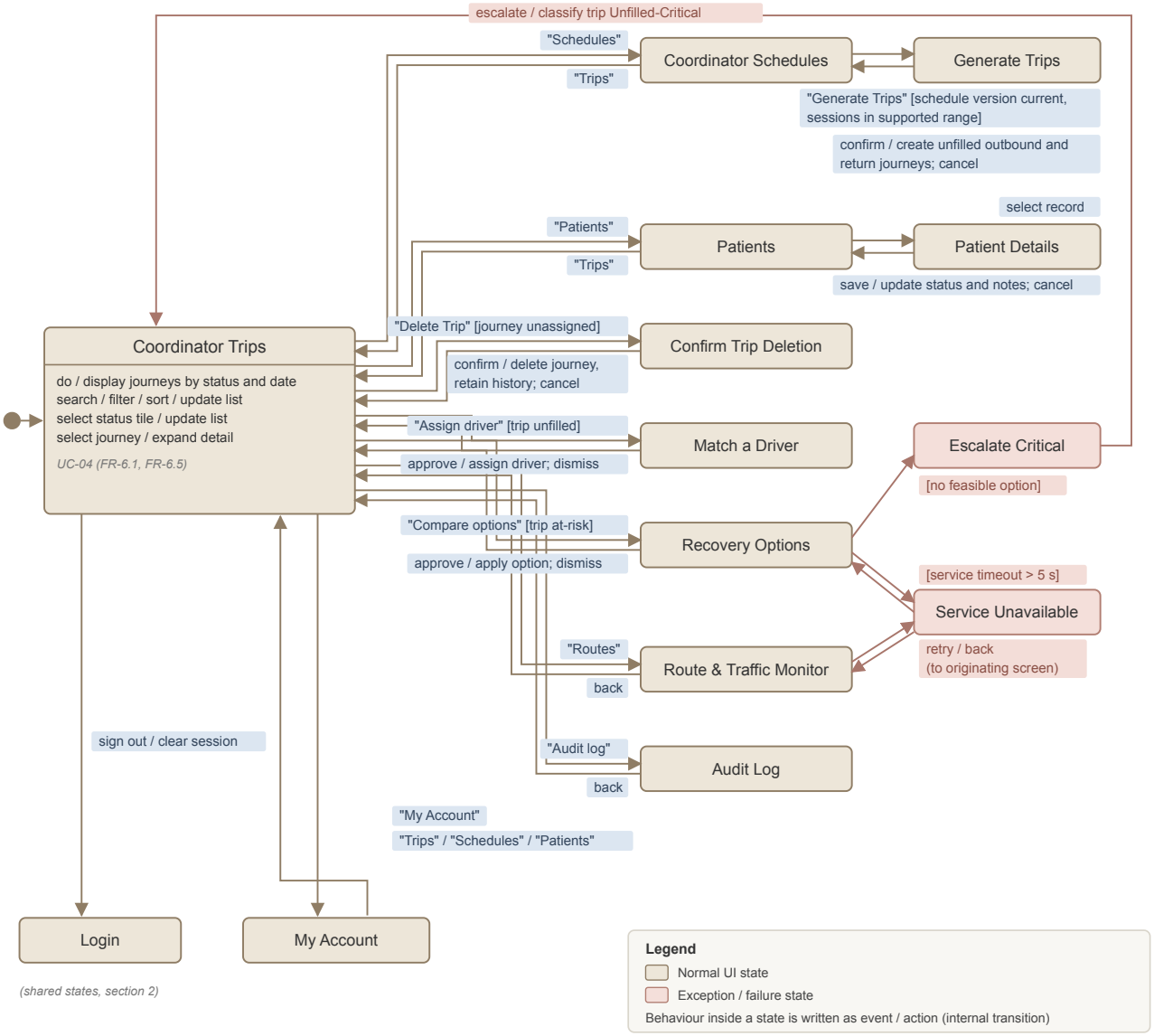
Dashed arrow: sign out from any function page of any role map returns here.
Login, Create Account, Complete Details, Account Inactive, Select Patient and My Account belong to no role map; every active role reaches My Account.
Coordinator accounts are provisioned manually, so no sign-up path enters the Coordinator map.

Design notes

- Coordinator Trips, Driver Trips and Patient Trips are the entry points into the three role maps in Sections 3–5. They are one route, `/dashboard/trips`, rendered for the role of the session; `/dashboard` only redirects to it, so the model carries no separate landing state.

- Authentication is Google OAuth, so Login carries no credential form. A failed sign-in is modelled as a **self-transition** on Login carrying the action `show inline error`. While the service health check fails, the sign-in control is disabled; that is a condition of the state, not a transition.
- **Create Account** and **Complete Details** are the two sign-up steps on the landing card. Next and Back only move between them and change no stored data; the account and, for a Patient, the owned profile are saved atomically on Sign Up. Only Patient, Caregiver and Driver can be chosen, so no sign-up path enters the Coordinator map — coordinators are provisioned manually. A Patient whose owned profile is incomplete enters Complete Details instead, whether that is detected at Login or later on My Account.
- **Account Inactive** is a state rather than an inline message. An authenticated account that is not active reaches a screen whose only exit is sign out, so an inactive account cannot proceed into a role map.
- **My Account** belongs to no role map: every active role reaches it from any function page and returns through the same navigation. A Patient edits their owned details in place on this screen, so edit, save and cancel are internal transitions, not a separate form state. Notifications are likewise a header panel available on every function page, not a state.
- Caregivers authenticate through the same Patient entry point using their own account. A Caregiver linked to at most one Patient transitions directly to Patient Trips, under the same guarded transition as a Patient; a Caregiver linked to more than one Patient passes through **Select Patient** first, since the system cannot determine the scope of the session without that choice. Caregiver linking (UC-02.1) is not implemented, so no Caregiver currently has a linked Patient and Select Patient is not yet reachable.
- Signing out from any function page returns to Login rather than to a final state, because the application does not terminate on sign-out. Each authenticated session is bound to a single role (FR-1.3), so only one role map is reachable per session.
- Related use case: UC-01 Authenticate User (FR-1.2, FR-1.3, NFR-5.1, NFR-5.2).

3. Coordinator



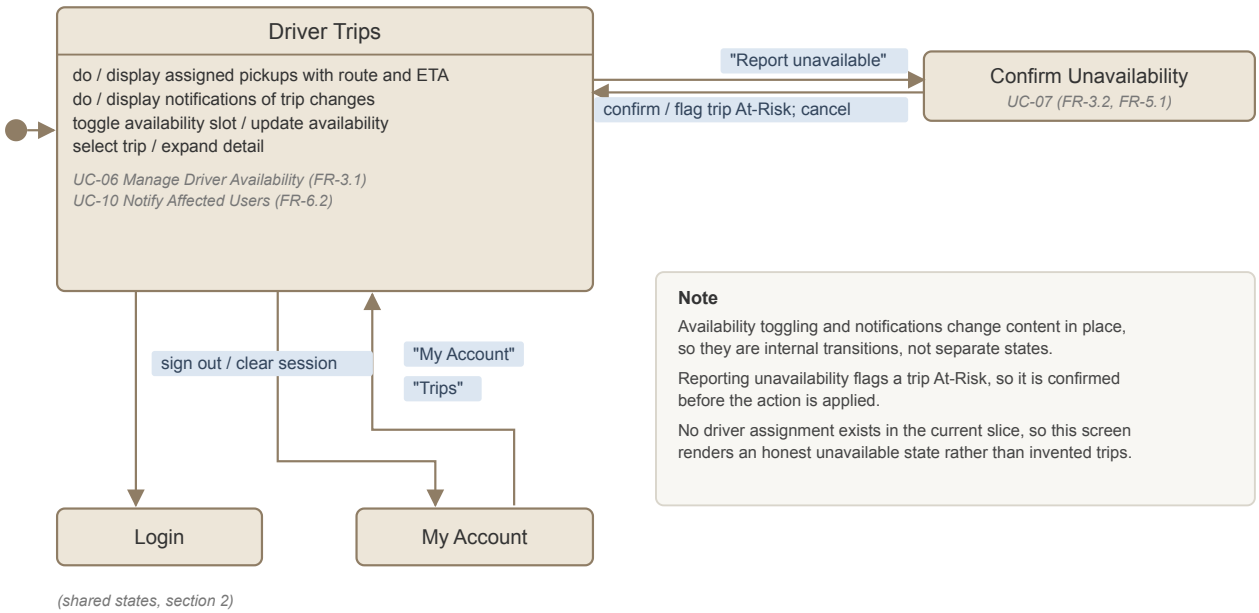
Design notes

- Coordinator Trips carries the UC-04 status categories. Its status tiles, search, filters, sorting, pagination and journey-details dialog all change content within the same screen, so they are internal transitions, not states. The same holds for the Patients and Coordinator Schedules lists.
- Guards are shown on the transitions: "Assign driver" is available only on Unfilled trips, "Compare options" only on At-Risk trips, and "Delete Trip" only on a journey with no driver assigned.
- **Coordinator Schedules** is read-only: treatment requirements are recorded by the Patient, not the coordinator. **Generate Trips** replaces the New Trip state of the earlier model. Journeys are derived from a selected treatment schedule over a bounded future range using the treatment times and the standard 15-minute buffer (UC-03), rather than typed in by hand, and generation is idempotent.
- **Patients** covers UC-02. Record creation belongs to sign-up and deletion is not permitted, so Patient Details carries neither an "add new" transition nor an archive action; a coordinator may save only Status and Notes.
- **Confirm Trip Deletion** is a state for the same reason as the other confirmations: removing a journey cannot be undone from the list. It applies only to an unassigned journey, and the journey's

history is retained.

- **Escalate Critical** covers FR-5.4. Where the system can generate no feasible recovery option, the trip is classified Unfilled-Critical and escalated for manual intervention rather than silently returning the coordinator to Coordinator Trips. The only exit is `escalate`, so a trip in this state cannot be left without being escalated.
- **Service Unavailable** covers the external-dependency failure implied by FR-4.1, FR-4.2 and NFR-2.1. Where routing or traffic data cannot be retrieved within the five-second bound, the coordinator is shown a recoverable failure state rather than stale or absent data. The same exception applies to Recovery Options, which depends on the same services; retry or back returns the coordinator to whichever of the two screens raised it.
- Match a Driver, Recovery Options, Route & Traffic Monitor and Audit Log are designed states with no data path behind them in the current slice; they are modelled here because the use cases require them, not because a screen exists.
- The Audit Log serves UC-04 (alternative flow: view audit log) and FR-6.4.
- Related use cases: UC-02 Manage Patient Records, UC-03 Manage Treatment and Transport Schedules, UC-04 Monitor Transport Operations, UC-05 Resolve At-Risk Trip, UC-08 Assess Route and ETA.

4. Driver



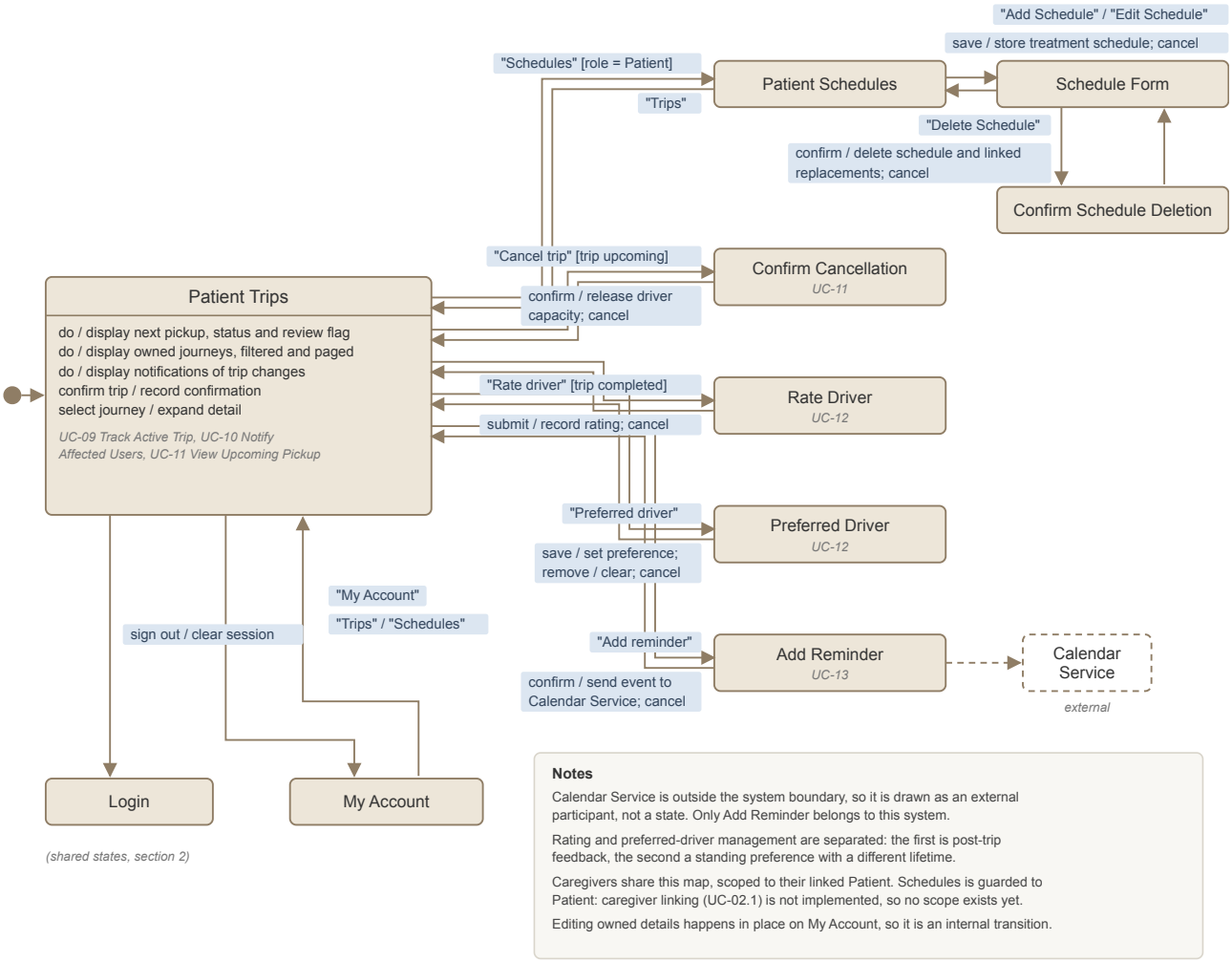
(shared states, section 2)

Design notes

- Driver Trips is the Trips function page rendered for a Driver: assigned pickups and weekly availability on one scrolling page, with no tab switch between them.
- No driver or vehicle assignment exists in the current slice, so the shipped screen is an honest unavailable state. The internal transitions above remain the designed UC-06 and UC-10 behaviour for when assignment exists.
- Toggling an availability slot updates the grid in place and is an internal transition (`toggle availability slot / update availability`), not a `do /` activity, because it is triggered by a user event and completes.

- Notifications appear in-page, in the shared header panel, so they are modelled as a display activity on this state rather than as a separate screen.
- **Confirm Unavailability** is modelled as a state. Reporting unavailability flags the trip At-Risk and requires coordinator intervention (FR-5.1), so it is confirmed before the action is applied rather than being a single irreversible tap.
- Related use cases: UC-06 Manage Driver Availability, UC-07 Report Driver Unavailability, UC-10 Notify Affected Users.

5. Patient & Caregiver



Design notes

- Patient Trips is a single scrolling page carrying the next pickup — its pickup status, journey date and direction and any review flag — above a date-filtered, paged list of owned journeys. Route addresses and the treatment-derived constraints are in that journey's details dialog, which is an internal transition. There is no separate tracking screen.
- **Patient Schedules** carries the guard `[role = Patient]`: a Caregiver is not routed to it until UC-02.1 linking exists. **Schedule Form** covers both adding and editing a treatment schedule; the whole-series / one-session scope choice inside it changes the same form, so it is an internal transition. **Confirm Schedule Deletion** is a state because deleting a series also deletes its linked replacements.

- Confirming an upcoming trip acts in place and is an internal transition. Cancelling does not: **Confirm Cancellation** is a state, because cancelling a dialysis journey releases driver capacity and cannot be undone by the patient. It carries the guard `[trip upcoming]`, since a completed trip cannot be cancelled.
- **Rate Driver** carries the guard `[trip completed AND role = Patient]`. A trip that has not taken place cannot be rated, and rating is the Patient's own: a Caregiver is not routed to it. **Preferred Driver** carries `[role = Patient]` for the same reason.
- Rating and preferred-driver management are modelled as two states rather than one. They have different triggers and different lifetimes: one is feedback on a completed trip, the other a standing preference that persists across trips. Both trace to UC-12.
- **Calendar Service** is outside the system boundary and is therefore drawn as an external participant, not as a state. The state belonging to this system is **Add Reminder**; the call to the external service is an action on the confirm transition.
- This map is shared by Patients and Caregivers under the login assumption in Section 2. A Caregiver's view is scoped to the linked Patient selected at sign-in, and a Caregiver may confirm or cancel trips on that Patient's behalf.
- Related use cases: UC-09 Track Active Trip, UC-10 Notify Affected Users, UC-11 View Upcoming Pickup, UC-12 Manage Driver Feedback and Preference, UC-13 Manage Calendar Reminders.

6. Transition Summary

The transitions below are the formal statement of the model; the diagrams are their visual form. A dash in the Event column denotes a transition triggered by its guard becoming true rather than by a user action.

From	Event	Guard	Action	To
Login	continue with Google	[account active, role = Coordinator]	—	Coordinator Trips
Login	continue with Google	[account active, role = Driver]	—	Driver Trips
Login	continue with Google	[account active, role = Patient, owned profile complete]	—	Patient Trips
Login	continue with Google	[role = Caregiver, linked patients ≤ 1]	—	Patient Trips
Login	continue with Google	[role = Caregiver, linked patients > 1]	—	Select Patient
Login	continue with Google	[no account record]	—	Create Account
Login	continue with Google	[role = Patient, owned profile incomplete]	—	Complete Details
Login	continue with Google	[account inactive]	—	Account Inactive
Login	continue with Google	[Google sign-in failed]	show inline error	Login (self)
Create Account	Next	—	—	Complete Details
Create Account	change Google account	—	restart Google selection	Create Account (self)
Create Account	Cancel	—	clear session	Login
Complete Details	Back	—	—	Create Account
Complete Details	Sign Up	[role = Patient]	save account and owned patient profile	Patient Trips
Complete Details	Sign Up	[role = Caregiver]	save account	Patient Trips
Complete Details	Sign Up	[role = Driver]	save account	Driver Trips
Complete Details	Cancel	—	clear session	Login
Account Inactive	sign out	—	clear session	Login
Select Patient	select linked patient	—	set session scope to patient	Patient Trips
any function page (all role maps)	"My Account"	—	—	My Account
My Account	"Trips"	[role = Coordinator]	—	Coordinator Trips
My Account	"Trips"	[role = Driver]	—	Driver Trips

From	Event	Guard	Action	To
My Account	"Trips"	[role = Patient OR Caregiver]	—	Patient Trips
My Account	"Schedules"	[role = Coordinator]	—	Coordinator Schedules
My Account	"Schedules"	[role = Patient]	—	Patient Schedules
My Account	"Patients"	[role = Coordinator]	—	Patients
My Account	edit	[role = Patient]	unlock owned details	My Account (self)
My Account	save	[role = Patient, version current]	update owned profile	My Account (self)
My Account	cancel	[role = Patient]	discard changes	My Account (self)
My Account	—	[role = Patient, owned profile missing]	—	Complete Details
Coordinator Trips / Patients	"Schedules"	—	—	Coordinator Schedules
Coordinator Schedules / Patients	"Trips"	—	—	Coordinator Trips
Coordinator Trips / Coordinator Schedules	"Patients"	—	—	Patients
Coordinator Schedules	"Generate Trips"	[schedule version current, sessions in supported range]	—	Generate Trips
Generate Trips	confirm	[future range ≤ 90 days]	create unfilled outbound and return journeys	Coordinator Schedules
Generate Trips	cancel	—	—	Coordinator Schedules
Coordinator Trips	"Delete Trip"	[journey unassigned]	—	Confirm Trip Deletion
Confirm Trip Deletion	confirm	[version current]	delete journey, retain history	Coordinator Trips
Confirm Trip Deletion	cancel	—	—	Coordinator Trips
Coordinator Trips	"Assign driver"	[trip unfilled]	—	Match a Driver
Match a Driver	approve	—	assign driver	Coordinator Trips
Match a Driver	dismiss	—	—	Coordinator Trips
Coordinator Trips	"Compare options"	[trip at-risk]	—	Recovery Options
Recovery Options	approve	—	apply recovery option	Coordinator Trips
Recovery Options	dismiss	—	—	Coordinator Trips
Recovery Options	—	[no feasible option]	—	Escalate Critical
Recovery Options	—	[service timeout > 5 s]	—	Service Unavailable

From	Event	Guard	Action	To
Escalate Critical	escalate	—	classify trip Unfilled-Critical	Coordinator Trips
Coordinator Trips	"Routes"	—	—	Route & Traffic Monitor
Route & Traffic Monitor	back	—	—	Coordinator Trips
Route & Traffic Monitor	—	[service timeout > 5 s]	—	Service Unavailable
Service Unavailable	retry / back	[entered from Route & Traffic Monitor]	—	Route & Traffic Monitor
Service Unavailable	retry / back	[entered from Recovery Options]	—	Recovery Options
Coordinator Trips	"Audit log"	—	—	Audit Log
Audit Log	back	—	—	Coordinator Trips
Patients	select record	—	—	Patient Details
Patient Details	save	[version current]	update status and notes	Patients
Patient Details	cancel	—	—	Patients
Driver Trips	"Report unavailable"	—	—	Confirm Unavailability
Confirm Unavailability	confirm	—	flag trip At-Risk	Driver Trips
Confirm Unavailability	cancel	—	—	Driver Trips
Patient Trips	"Schedules"	[role = Patient]	—	Patient Schedules
Patient Schedules	"Trips"	—	—	Patient Trips
Patient Schedules	"Add Schedule" / "Edit Schedule"	—	—	Schedule Form
Schedule Form	save	[version current, no overlapping session]	store treatment schedule	Patient Schedules
Schedule Form	cancel	—	—	Patient Schedules
Schedule Form	"Delete Schedule"	[schedule owned]	—	Confirm Schedule Deletion
Confirm Schedule Deletion	confirm	—	delete schedule and linked replacements	Patient Schedules
Confirm Schedule Deletion	cancel	—	—	Schedule Form
Patient Trips	"Cancel trip"	[trip upcoming]	—	Confirm Cancellation
Confirm Cancellation	confirm	—	release driver capacity	Patient Trips
Confirm Cancellation	cancel	—	—	Patient Trips

From	Event	Guard	Action	To
Patient Trips	"Rate driver"	[trip completed AND role = Patient]	—	Rate Driver
Rate Driver	submit	—	record rating	Patient Trips
Rate Driver	cancel	—	—	Patient Trips
Patient Trips	"Preferred driver"	[role = Patient]	—	Preferred Driver
Preferred Driver	save	—	set preferred driver	Patient Trips
Preferred Driver	remove	—	clear preferred driver	Patient Trips
Preferred Driver	cancel	—	—	Patient Trips
Patient Trips	"Add reminder"	—	—	Add Reminder
Add Reminder	confirm	—	send event to Calendar Service	Patient Trips
Add Reminder	cancel	—	—	Patient Trips
any function page (all role maps)	sign out	—	clear session	Login